$$3y^2\frac{dy}{dx} = 3x^2 \implies \frac{dy}{dx} = \frac{x^2}{y^2}$$

At $(-a, b)$, $\dfrac{dy}{dx} = \dfrac{a^2}{b^2}$

Tangent: $y - b = \dfrac{a^2}{b^2}(x + a) \implies b^2y - a^2x = a^3 + b^3$

When $a = 1,\ b = 2 \implies y^3 = x^3 + 9$

Tangent: $4y - x = 9$

$$\implies \left(\frac{x+9}{4}\right)^3 = x^3 + 9$$

$$x^3 + 27x^2 + 243x + 729 = 64x^3 + 576$$

$$\therefore 7x^3 - 3x^2 - 27x - 17 = 0$$

$$7x^3 - 3x^2 - 27x - 17 = (x+1)^2(7x - 17)$$

When $x = \dfrac{17}{7} \implies y = \dfrac{20}{7}$

$$\because y^3 = x^3 + 1^3 + 2^3$$

$$\therefore \left(\frac{20}{7}\right)^3 = \left(\frac{17}{7}\right)^3 + 1^3 + 2^3$$

$$\therefore 20^3 = 17^3 + 7^3 + 14^3$$

$$\implies \boxed{p = 20, \quad q = 17, \quad r = 7, \quad s = 14}$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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